

Ben Greenman
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Kahlert School of Computing
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RESEARCH INTERESTS

General interests: Language design issues regarding proofs, performance, and people. What guarantees do languages offer, how efficiently do they run, and to what extent do they help users meet their goals?

Keywords: Gradual typing, Language design, Formal methods, Human factors

EMPLOYMENT AND EDUCATION

- Assistant Professor of Computer Science, University of Utah July 2023 – ongoing
- Postdoctoral Researcher, Brown University 2021 – 2023
supported by the *CIFellows 2020* program
Mentor: *Shriram Krishnamurthi*
- Ph.D. in Computer Science, Northeastern University 2014 – 2020
Advisor: *Matthias Felleisen*
Thesis: *Deep and Shallow Types*
- M. Eng. in Computer Science, Cornell University 2013 – 2014
Advisor: *Ross Tate*
- Programmer, Rentenna Inc. 2012 – 2014
- B.S. in Industrial and Labor Relations (ILR), Cornell University 2010 – 2013
Minor in Computer Science
- General studies, Hudson Valley Community College 2009 – 2010
toward a guaranteed transfer to Cornell ILR

HONORS AND AWARDS

- Open Source Research Experience: Type Narrowing: A Language Design Benchmark 2025
received summer support for Siva Sathyaseelan, an undergraduate researcher from IIT (BHU) Varanasi
sponsored by the *NSF 2025 Summer of Reproducibility*

- **Open Source Research Experience: Static Python Perf** 2024
received summer support for Mrigank Pawagi, an undergraduate researcher from IIS Bengaluru sponsored by the NSF 2024 Summer of Reproducibility
- **CRA/CCC/NSF CI Fellowship** 2021 – 2023
- **SIGPLAN Student Scholarship to 50 Years of the ACM A.M. Turing Award** 2017
- **Northeastern CCIS Graduate Community Service Award** 2016
- **Cornell CS Teaching Award** 2014, 2013
- **Distinguished Paper Award** Programming 2026, CAV 2025, ECOOP 2025, Programming 2023
- **Distinguished Artifact Award** ECOOP 2025

FUNDING

- **Price College: AI, Innovation, and Student Success** May 2026
PI Wiese, Co-PIs Greenman, Kopta
- **NSF CS2: On-Demand Semantic Analysis via Program Synthesis** April 2026
PI Greenman, Co-PIs Regehr, Shankar, D’Antoni (UCSD) **IIS-2546821**
- **Price College VPR Seed Grant Competition** 2025

PUBLICATIONS

JOURNAL

- Nathaniel Hejduk, Ben Greenman, Matthias Felleisen, and Christos Dimoulas **TOPLAS 2025**
Navigating Mixed-Typed Migration with Profilers
- Ben Greenman, Christos Dimoulas, and Matthias Felleisen. **TOPLAS 2023**
Typed–Untyped Interactions: A Comparative Analysis
- Ben Greenman, Asumu Takikawa, Max S. New, Daniel Feltey, Robert Bruce Findler, Jan Vitek, and Matthias Felleisen. **JFP 2019**
How to Evaluate the Performance of Gradual Type Systems

CONFERENCE & SYMPOSIUM

- Dominic Kennedy, Xuanyu Peng, Yuyou Fan, John Regehr, Loris D’Antoni, Ben Greenman **POPL 2027**
A Fistful of Abstract Values: Synthesizing the Transformers Compilers Actually Need

- Ali Ghanbari, Ben Greenman, Sasan Tavakkol, Shibbir Ahmed ISSTA 2026
Provably Lossless Acceleration of DNN Mutation Testing via Memoization
- Xuanyu Peng, Dominic Kennedy, Yuyou Fan, Ben Greenman, John Regehr, Loris D’Antoni POPL 2026
Nice to Meet You: Synthesizing Practical Abstract Transformers
- Ashton Wiersdorf and Ben Greenman Programming 10.3, 2026
Chorex: Restartable, Language-Integrated Choreographies
- Hanwen Guo and Ben Greenman Programming 10.2, 2026
If-T: A Benchmark for Type Narrowing
Editors’ Choice Award
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi CAV 2025
A Misconception-Driven Adaptive Tutor for Linear Temporal Logic
Distinguished Paper Award
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi ECOOP 2025
Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design
Distinguished Paper Award
- Ashton Wiersdorf, Stephen Chang, Matthias Felleisen, and Ben Greenman ECOOP 2024
Type Tailoring
- Ben Greenman, Siddhartha Prasad, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, and Milda Zizyte FM 2024
Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic
- Tim Nelson, Ben Greenman, Siddhartha Prasad, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, and Shriram Krishnamurthi OOPSLA 2024
Forge: A Tool and Language for Teaching Formal Methods
- Ben Greenman, Alan Jeffrey, Shriram Krishnamurthi, and Mitesh Shah Programming 8.3, 2024
Privacy-Respecting Type Error Telemetry at Scale
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi Programming 8.2, 2024
Conceptual Mutation Testing for Student Programming Misconceptions
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi CompEd, December 2023
Generating Programs Trivially: Student Use of Large Language Models
- Ben Greenman, Matthias Felleisen, and Christos Dimoulas OOPSLA 2023
How Profilers Can Help Navigate Type Migration
- Matthew Flatt, Taylor Allred, Nia Angle, Stephen De Gabrielle, Robert Findler, Jack Firth, Kiran Gopinathan, Ben Greenman, Siddhartha Kasivajhula, Alex Knauth, Jay McCarthy, Sam Phillips, Sorawee Porncharoenwase, Jens Axel Sogaard, and Sam Tobin-Hochstadt OOPSLA 2023
Rhombus: A New Spin on Macros Without All The Parentheses

- Lukas Lazarek, Ben Greenman, Matthias Felleisen, and Christos Dimoulas ICFP 2023
How to Evaluate Blame for Gradual Types, Part 2
- Ben Greenman ACM REP, June 2023
GTP Benchmarks for Gradual Typing Performance
- Ben Greenman, Sam Saarinen, Tim Nelson, and Shriram Krishnamurthi Programming 7.2, 2023
Little Tricky Logic: Misconceptions in the Understanding of LTL
- Kuang-Chen Lu, Ben Greenman, Carl Meyer, Dino Viehland, Aniket Panse, and Shriram Krishnamurthi Programming 7.1, 2023
Gradual Soundness: Lessons from Static Python
- Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, and Shriram Krishnamurthi Koli Calling 2022
Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions
- Ben Greenman PLDI 2022
Deep and Shallow Types for Gradual Languages
- Ben Greenman, Lukas Lazarek, Christos Dimoulas, and Matthias Felleisen Programming 6.2, 2022
A Transient Semantics for Typed Racket
- Kuang-Chen Lu, Ben Greenman, and Shriram Krishnamurthi Programming 6.2, 2022
Types for Tables: A Language Design Benchmark
Editors' Choice Award
- Lukas Lazarek, Ben Greenman, Matthias Felleisen, and Christos Dimoulas ICFP 2021
How to Evaluate Blame for Gradual Types
- Ben Greenman, Matthias Felleisen, and Christos Dimoulas OOPSLA 2019
Complete Monitors for Gradual Types
- Preston Tunnell Wilson, Ben Greenman, Justin Pombrio, Shriram Krishnamurthi DLS 2018
The Behavior of Gradual Types: A User Study
- Daniel Feltey, Ben Greenman, Christophe Scholliers, Robert Bruce Findler, and Vincent St. Amour OOPSLA 2018
Collapsible Contracts: Fixing a Pathology of Gradual Typing
- Ben Greenman, Matthias Felleisen ICFP 2018
A Spectrum of Type Soundness and Performance
- Ben Greenman, Zeina Migeed PEPM 2018
On the Cost of Type-Tag Soundness
- Sam Tobin-Hochstadt, Matthias Felleisen, Robert Bruce Findler, Matthew Flatt, Ben Greenman, Andrew M. Kent, Vincent St-Amour, T. Stephen Strickland, and Asumu Takikawa SNAPL 2017
Migratory Typing: 10 Years Later
- Stephen Chang, Ben Greenman, and Alex Knauth POPL 2017
Type Systems as Macros

- Asumu Takikawa, Daniel Feltey, Ben Greenman, Max S. New, Jan Vitek, and Matthias Felleisen
Is Sound Gradual Typing Dead? POPL 2016
- Ben Greenman, Fabian Muehlboeck, and Ross Tate
Getting F-Bounded Polymorphism into Shape PLDI 2014

WORKSHOP

- Shriram Krishnamurthi, Siddhartha Prasad, Tim Nelson, and Ben Greenman
A Tutor for Linear Temporal Logic TEAL@FLOC 2026
- Ashton Wiersdorf and Ben Greenman
Bushwhacking Toward Event-Driven Chorex Choreo 2026
- Dibri Nsofor and Ben Greenman
Toward a Corpus Study of the Dynamic Gradual Type HATRA 2024
- Taylor Allred, Xinyi Li, Ashton Wiersdorf, Ben Greenman, and Ganesh Gopalakrishnan
FlowFPX: Nimble Tools for Debugging Floating-Point Exceptions JuliaCon 2023
- Asumu Takikawa, Daniel Feltey, Ben Greenman, Max S. New, Jan Vitek, and Matthias Felleisen.
Position Paper: Performance Evaluation for Gradual Typing STOP 2015

INVITED TALKS

- **Formally Speaking Seminar at Code Metal**
Typing Python, Gradually September 2026
- **WITS**
Type Narrowing the Hard Way January 2026
- Amazon Compilers Tech Talk
Nice to Meet You: Synthesizing Practical MLIR Transformers December 2025
- RPI CS Seminar
Kicking the Ladder Away: From Gradual Types to Plain Types June 2025
- **Iowa State CS Colloquium**
Toward a Science of Type System Design November 2024
- **Research Challenges in Computing @ University of Utah**
Rigorous Methods for Language Design 2024
- **PLT @ Northwestern University**
Teaching Formal Methods with Forge September 2024
- **IETF 120: Usable Formal Methods Research Group**
Forge: Usable Model-Finding July 2024

- [BYU Grad Seminar](#) November 2023
How Profilers Can Help Navigate Type Migration
- [TLf@AAAI-SSS'23](#) March 2023
Towards LTLf Misconceptions
- [VardiFest](#) 2022
[NJPLS](#)
Little Tricky Logic: Misconceptions in the Understanding of LTL
- [Racket Con](#) 2020, 2022
Shallow Typed Racket
Shallow and Optional Types for Typed Racket
- [Boston University POPV Seminar](#) 2020
Complete Monitoring for Gradual Types
- [GRACE Workshop](#) 2018
Three Approaches to Gradual Typing

TEACHING

UTAH

			Enrollment (Responded)	Course (Avg)	Instructor (Avg)
Fall 26	CS 4470	Compilers	20 (??)	<i>TBD (TBD)</i>	<i>TBD (TBD)</i>
Spring 25	CS 5110/6110	Software Verification	46 (38)	5.09 (5.09)	5.74 (5.20)
	CS 7936	Ph.D Seminar	5 (1)	6 (5.09)	6 (5.20)
Fall 25	COMP 1020	Programming for All 2	68 (43)	5.42 (5.17)	5.70 (5.31)
Spring 25	CS 4470	Compilers	58 (51)	5.28 (5.24)	5.43 (5.28)
	CS 7936	Ph.D Seminar	6 (1)	6	6
Fall 24	<i>N/A</i>	<i>parental leave</i>			
Spring 24	CS 5110/6110	Software Verification	22 (20)	5.5 / 5.82 (5.18)	6 / 5.68 (5.21)
Fall 23	CS 3520/6520	Programming Languages	159 (77)	5.32 / 5.82 (5.12)	5.45 / 5.68 (5.19)

BROWN

- Topics in PL and Systems: Tables and Humans 2021
Seminar Organizer & Scribe

NORTHEASTERN

- Software Development 2018, 2020
Teaching Assistant
- Fundamentals I 2016
Teaching Assistant

- Object-Oriented Design 2016
Teaching Assistant

CORNELL

- Functional Programming and Data Structures 2012 – 2014
Teaching Assistant

ADVISING

PH.D.

- Ashton Wiersdorf, started Fall 2023
- Dominic Kennedy, started Fall 2024
- Hanwen Guo, started Fall 2024

MASTERS

- Jackson Brough, BS expected Spring 2027
Thesis: *Univalent Real Analysis using Locators*
- Dibri Nsofor, MSc Fall 2023 – Spring 2026
Project: *Data Science for Gradual Types*
- Suyasha Bobhate, IS Fall 2023 graduated Spring 2024
Project: *Quantum Key-Value Stores*

UNDERGRAD

- Riley Denoyer, BS expected Spring 2027
Project: *Clay: Borrow Checking without the Unique-Writer Restriction*
- Robert Morelli, BS expected Spring 2027
Project: *Automatic Cast Insertion for Progressive Gradual Types*
- Jackson Brough, BS Fall 2025 – Spring 2026
Thesis: *Formalizing the Real Numbers in Homotopy Type Theory with Cubical Agda* ([arxiv 2604.24782](#))

COMMITTEE MEMBERSHIP

- [Guillaume Duboc](#), Ph.D, advisor [Giuseppe Castagna](#)
- Zhaofeng Li, Ph.D, advisor [Anton Burtsev](#)
- Sara Nurollahian, Ph.D, advisor [Eliane Wiese](#)

INFORMAL MENTEES

Siva Sathyaseelan		IIT (BHU) Varanasi	Summer 2025
Mrigank Pawagi		IIS Bengaluru	Summer 2024
Vivaan Rajesh		Hillcrest High School	2023 – 2024
Siddhartha Prasad	Ph.D.	Brown University	2022 – ongoing
Rob Durst			Fall 2023
Caspar Popova			Spring – Fall 2023
Aniket Karna	M.S.	University of Utah	Fall 2023
Taylor Allred	M.S.	University of Utah	2022 – 2023
Qianfan Chen	Sc.B.	Brown University [thesis]	2021 – 2022
Kuang-Chen Lu	Ph.D.	Brown University	2021 – 2022
Milo Davis	B.S.	Northeastern University	2017
Zeina Migeed	B.S.	Northeastern University	2016 – 2017

DEPARTMENT, COLLEGE, AND UNIVERSITY SERVICE

- Committee Member: Lecturing Faculty Hiring Fall 2025 – Spring 2026
- Faculty Mentor: CS 1960: Success in Computing Summer 2025 – ongoing
- Committee Member: Graduate Admissions Spring 2026, 2025
- Teacher: Price College Hi-Gear Summer Camp Summer 2026, 2025
- Teacher: Price College Exploring Engineering Summer Camp Summer 2024
- Teaching Area Chair: Programming Languages and Web Fall 2023 – ongoing
- Committee Member: K-12 Outreach Planning Committee Fall 2023 – Summer 2025

EXTERNAL SERVICE

- Co-Chair of Workshop Organization ICFP 2026, ICFP/SPLASH 2025
- Co-Chair of Artifact Evaluation Committee & ERC OOPSLA 2023, 2022
- Program Committee
 - DLS 2022
 - HATRA 2025, 2024, 2023, 2022
 - ICFP 2026, 2021
 - OOPSLA 2025
 - POPL 2027
 - PLDI 2025, 2021
 - Scheme 2025
 - SOAP 2024
 - TFP 2026, 2025, 2023
- External Review Committee ESOP 2023, ICFP 2023, OOPSLA 2026
- Journal Review
 - JFP 2026, 2024, 2023, 2020, 2019
 - JuliaCon 2024

- SoftwareX 2025
- STTT 2024
- TOPLAS 2025, 2023
- NSF Panel Review 2025, 2024
- Artifact Evaluation Committee ECOOP 2017; OOPSLA 2017, 2016
- Session Chair ICFP 2026, 2021; NJPLS 2023; OOPSLA 2023; POPL 2026; Programming 2026
- SIGPLAN-M Long-Term Mentor Fall 2024 – Fall 2026
- El Turco: Human-AI dialogue Spring 2024
show: Mori Art Museum, 2025-02-13 – 2025-06-08
- Senior Division Judge: University of Utah Science and Engineering Fair (USEF) 2026, 2025
- Participant: Higher Ed Review of AP Computer Science A Exam 2026

PROFESSIONAL MEMBERSHIPS

- IEEE, Member 2023 – ongoing
- IEEE Computer Society, Member 2023 – ongoing
- ACM, Member 2023 – ongoing
- ACM SIGPLAN, Member 2016 – ongoing
- Sigma Xi, Member 2025 – ongoing
The Scientific Research Honor Society
- Phi Theta Kappa, Member 2013
2-year college Honor Society

BIOGRAPHY

Ben Greenman is an assistant professor in the Kahlert School of Computing at the University of Utah. He earned his PhD from Northeastern University in 2020 under the supervision of Matthias Felleisen. He was a CIFellows 2020 postdoc at Brown University mentored by Shriram Krishnamurthi. His research seeks to elevate language design from an art to a science by systematic methods that assess formal guarantees, performance, and human factors concerns.